

# Chapter 2: The Invisible Living World: Beyond Our Naked Eye

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Q1. Have you ever wondered what you might see if the invisible world around you became visible?

**Answer:** If the invisible microscopic world became visible to the naked eye, we would see millions of tiny microorganisms (bacteria, fungi, protozoa, and viruses) covering almost every surface around us—floating in the air, swimming in water, living in the soil, on our skin, and inside our gut . We would observe them breaking down waste, interacting with each other, and performing vital biological processes essential for maintaining ecological balance on Earth .

Q2. How do you think your observation of this hidden world might change the way you think about size, complexity, or even what counts as 'living'?

**Answer:** **Size:** It reveals that life exists on vastly different scales—from microscopic single-celled organisms measuring a few micrometers to massive multicellular animals .

**Complexity:** It demonstrates that even a single microscopic cell is not a simple bag of liquid, but a complex, highly organized unit capable of respiration, growth, nutrition, and reproduction.

**What Counts as 'Living':** It expands our understanding of life beyond large, visible plants and animals to include single-celled organisms (like Amoeba and bacteria) and acellular entities (like viruses, which reproduce inside host cells).

Q3. Have you thought how these tiny living beings interact with each other?

**Answer:** Microorganisms interact in several ways:

**Symbiosis / Mutualism:** Rhizobium bacteria live in legume root nodules, providing fixed nitrogen to plants while receiving food and shelter in return .

**Decomposition:** Fungi and bacteria cooperate to break down dead organic matter, recycling nutrients back into the soil .

**Competition & Pathogenesis:** Some microbes compete for food and space, while others infect host organisms and cause diseases.

Q1. Various parts of a cell are given below.

Write them in the appropriate places in the diagram.

Given parts: Nucleus, Chloroplast, Cytoplasm, Cell wall, Cell membrane, Nucleoid.

Common to All Three Cells (Plant, Animal, & Bacterial): Cell membrane, Cytoplasm.

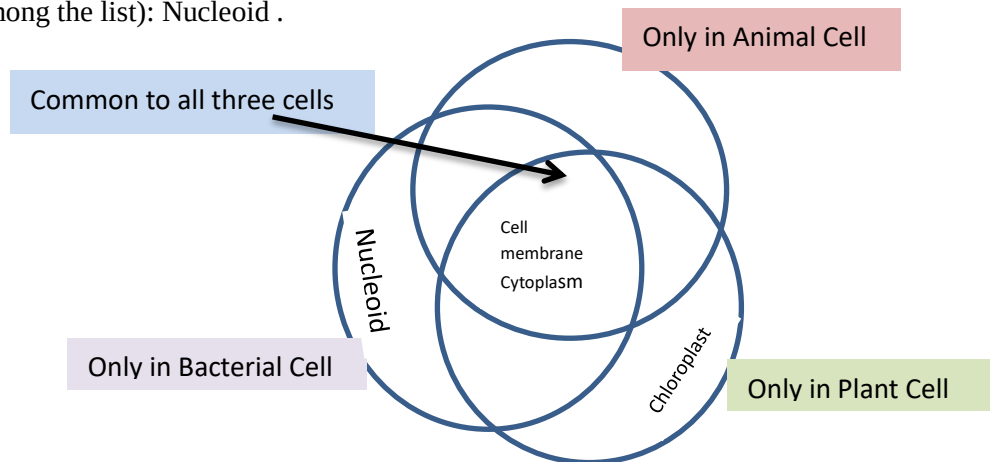
Common to Plant and Animal Cells (Absent in Bacterial Cell): Nucleus.

Common to Plant and Bacterial Cells (Absent in Animal Cell): Cell wall.

Only in Plant Cell (among the list): Chloroplast.

Only in Bacterial Cell (among the list): Nucleoid .

Ans.



Q2. Aanandi took two test tubes marked A and B. She put two spoonfuls of sugar solution in each. In test tube B, she added a spoonful of yeast.

Then she attached two incompletely inflated balloons to the mouth of each test tube and kept the set-up in a warm place.

(i) What do you predict will happen after 3–4 hours? She observed that the balloon attached to test tube B was inflated.

What can be a possible explanation for this?

Options:

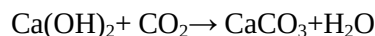
- (a) Water evaporated in test tube B and filled the balloon with water vapour.
- (b) The warm atmosphere expanded the air inside test tube B, which inflated the balloon.
- (c) Yeast produced a gas inside test tube B which inflated the balloon.
- (d) Sugar reacted with warm air, which produced gas, eventually inflating the balloon.

Correct Answer: (c) Yeast produced a gas inside test tube B which inflated the balloon.

Explanation: Yeast respires anaerobically by breaking down sugar (fermentation), releasing carbon dioxide gas ( $\text{CO}_2$ ), which collects in the balloon and causes it to inflate .

(ii) She took another test tube,  $\frac{1}{4}$  filled with lime water. She removed the balloon from test tube B in such a manner that the gas inside the balloon did not escape. She attached the balloon to the test tube with lime water and shook it well. What do you think she wants to find out?

Answer: She wants to confirm that the gas produced by the yeast during respiration is Carbon Dioxide (CO<sub>2</sub>). Carbon dioxide reacts with lime water calcium hydroxide, Ca(OH)<sub>2</sub> to turn it milky / cloudy due to the formation of insoluble calcium carbonate CaCO<sub>3</sub>.



Q3. A farmer was planting wheat crops in his field and added nitrogen-rich fertiliser. In the neighbouring field, another farmer was growing bean crops, but preferred not to add nitrogen fertiliser. Can you think of the reasons?

Answer: Bean crops are leguminous plants. Their root nodules harbor symbiotic Rhizobium bacteria, which capture atmospheric nitrogen gas and convert it into water-soluble nitrogenous compounds usable by the plant. Hence, leguminous crops enrich the soil naturally and do not require synthetic nitrogen fertilisers.

Wheat crops are non-leguminous plants and lack Rhizobium in their roots. They cannot directly fix atmospheric nitrogen, so the farmer must apply external nitrogen-rich fertilizers to ensure healthy crop yield.

Q4. Snehal dug two pits, A and B, in her garden. In pit A, she put fruit and vegetable peels mixed with dried leaves. In pit B, she dumped the same kind of waste without mixing it with dried leaves. She covered both pits with soil and observed after 3 weeks. What is she trying to test?

Answer: Objective: She is testing the effect of organic mixture composition (specifically adding carbon-rich dry leaves) on the rate and efficiency of composting/decomposition by soil microorganisms.

Scientific Basis: Dry leaves provide carbon, while fruit/vegetable peels provide nitrogen and moisture. A balanced Carbon-to-Nitrogen (C:N) ratio along with proper aeration (provided by dry leaves creating air pockets) accelerates the growth of decomposing bacteria and fungi, turning waste into nutrient-rich manure faster in Pit A than in Pit B.

Q5. Identify the following microorganisms:

(i) I live in every kind of environment, and inside your gut.

Answer: Bacteria (e.g., gut(पेट या अँत) bacteria like Lactobacillus or E. coli) .

(ii) I make bread and cakes soft and fluffy.

Answer: Yeast (a single-celled fungus) .

(iii) I live in the roots of pulse crops and provide nutrients for their growth.

Answer: Rhizobium bacteria.

Q6. Design an experiment to test that microorganisms need optimal temperature, air, and moisture for their growth.

### **Experimental Design:**

Materials: 4 slices of fresh bread, 4 sealed ziplock bags, water dropper, refrigerator, warm incubator/room.

### **Procedure:**

**Condition 1** (Control - Optimal conditions): Sprinkle a few drops of water on Slice 1 (moisture), seal it in a bag containing air, and place it in a warm dark room (25

**Condition 2** (Testing Low Temperature): Sprinkle water on Slice 2, seal with air, but keep it in a refrigerator.

**Condition 3** (Testing Lack of Moisture): Keep Slice 3 completely dry, seal with air, and store in a warm dark room.

**Condition 4** (Testing Lack of Air): Sprinkle water on Slice 4, vacuum-seal the bag completely to remove all air, and store in a warm dark room.

### **Observation after 4–5 days:**

Slice 1 (Warm + Moist + Air): Shows dense cottony growth of mould (fungi).

Slices 2, 3, and 4 show little to no fungal growth.

**Conclusion:** Microorganisms require optimal temperature, moisture, and air for growth.

Q7. Take 2 slices of bread. Place one slice in a plate near the sink. Place the other slice in the refrigerator. Compare after three days.

Note your observations and give reasons.

### **Observations:**

Bread near sink: Shows greenish-black cottony spots (bread mould/fungi) and gives off a stale odor.

Bread in refrigerator: Remains clean, fresh, and free from visible mould growth.

### **Reasons:**

The area near the sink provides high moisture/humidity and a warm room temperature, creating ideal conditions for airborne fungal spores to germinate and grow.

The low temperature inside the refrigerator inhibits microbial metabolic activity and growth, preserving the bread.

Q8. A student observes that when curd is left out for a day, it becomes more sour. What can be two possible explanations for this observation?

### Explanations:

**Rapid Bacterial Multiplication:** At warm room temperature, *Lactobacillus* bacteria multiply at a much faster rate compared to refrigerated conditions.

**Increased Lactic Acid Production:** As *Lactobacillus* continues to consume lactose sugar present in the milk/curd, it converts more lactose into lactic acid; the accumulation of lactic acid increases acidity, making the curd noticeably more sour.

Q9. Observe the set-up given in Fig. 2.15 (Flask A: Warm sugar solution + Yeast; Test tube B: Lime water) and answer the following questions:

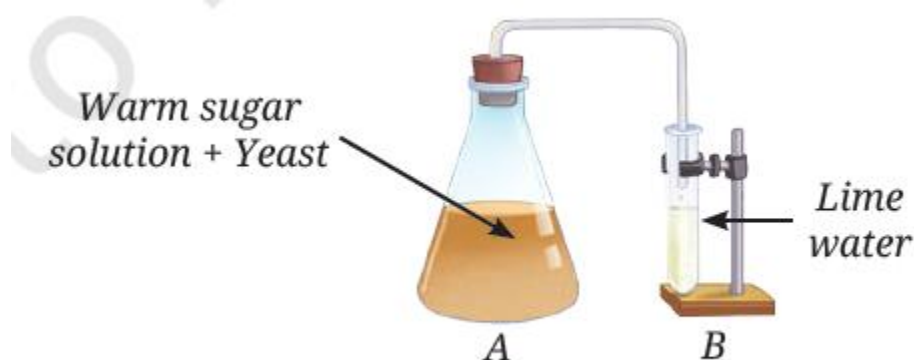


Fig. 2.15 : Experimental set-up

(i) What happens to the sugar solution in flask A?

Answer: Yeast undergoes anaerobic respiration (fermentation) in Flask A. It breaks down sugar into alcohol and carbon dioxide gas ( $\text{CO}_2$ ), causing the solution to froth/bubble and produce a characteristic alcoholic smell.

(ii) What do you observe in test tube B after four hours?

Why do you think this happened?

Answer:

Observation: The clear lime water in test tube B turns milky / cloudy white.

Reason: Carbon dioxide gas ( $\text{CO}_2$ ) produced by yeast in Flask A travels through the glass delivery tube into test tube B and reacts with lime water to form insoluble calcium carbonate.

(iii) What would happen if yeast was not added in flask A?

Answer: If yeast is omitted, no fermentation occurs. No carbon dioxide gas will be generated, so no gas bubbles will travel to test tube B, and the lime water in test tube B will remain clear.